

Missing Roots and Sign Errors in Quadratic Solutions

Answer Key

Algebra 1

Quick Answers

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|---------------------|---------------------------------------|
| 1. $-7, 7$ | 5. $-5, 3$ |
| 2. $(b) = -4, 4, —$ | 6. $1, 5$ |
| 3. $0, 5$ | 7. 3.19 |
| 4. $(b) = 0, 6, —$ | 8. $(b) = -3, 1 \cdot \frac{1}{2}, —$ |
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What is verified

5 of 8 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problems 2, 4, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Algebra 1

HSA-REI.B.4 – problems 1, 2, 3, 4, 5, 6, 7, 8

Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

1. If they got 7: reported only the positive square root
 2. If they got 4: kept only the positive square root
 3. If they got 5: divided both sides by x , discarding the root $x = 0$
 4. If they got 6: divided both sides by x , discarding the root $x = 0$
 5. If they got $-3, 5$: swapped the signs of the factor pair, factoring as $(x+3)(x-5)$
 6. If they got $-1, -5$: wrote b instead of $-b$ at the front of the numerator
 7. If they got -0.31 : dropped the sign of b and used -7 at the front of the numerator
 8. If they got $1, -6$: divided by 2 instead of $2a$
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1. Solve, and list every solution: $x^2 = 49$

Solution: Taking the square root of both sides produces two values, because $(-7)^2$ and 7^2 both equal 49.

$$x^2 = 49 \implies x = \pm\sqrt{49}$$

$$\boxed{x = -7 \quad \text{or} \quad x = 7}$$

2. Dana solves $2x^2 - 32 = 0$ like this: “ $2x^2 = 32$, so $x^2 = 16$, so $x = 4$.” (a) Name the mistake. (b) Write the complete solution set.

(a) — instructor-judged. Every line of Dana’s arithmetic is right; the error is in the *last step*. Taking a square root of both sides yields two values, $+4$ and -4 , and Dana kept only the positive one, so the solution set is incomplete. A response that says “the arithmetic is wrong” has not identified the error — -4 genuinely is a solution.

Solution (b):

$$\begin{aligned}2x^2 - 32 &= 0 \\x^2 &= 16 \\x &= \pm 4\end{aligned}$$

Check: $2(-4)^2 - 32 = 32 - 32 = 0$, so -4 really is a solution.

$$\boxed{x = -4 \quad \text{or} \quad x = 4}$$

3. Solve, and list every solution: $x^2 - 5x = 0$

Solution: There is a common factor of x . Factor it out and use the zero-product property — do *not* divide by x , because x may be zero.

$$\begin{aligned}x^2 - 5x &= 0 \\x(x - 5) &= 0 \\x = 0 \quad \text{or} \quad x - 5 &= 0\end{aligned}$$

$$\boxed{x = 0 \quad \text{or} \quad x = 5}$$

4. Marco solves $x^2 = 6x$ by dividing both sides by x , giving $x = 6$. (a) Name the error and say why it costs him a solution. (b) Write the complete solution set.

(a) — instructor-judged. Dividing both sides by x is only legal when $x \neq 0$. Here $x = 0$ satisfies the original equation ($0^2 = 6 \cdot 0$), so Marco’s division deleted a genuine solution. Full credit requires naming the lost root $x = 0$, not just saying “you cannot divide by a variable”.

Solution (b): Move everything to one side and factor instead of dividing.

$$\begin{aligned}x^2 &= 6x \\x^2 - 6x &= 0 \\x(x - 6) &= 0\end{aligned}$$

$$\boxed{x = 0 \quad \text{or} \quad x = 6}$$

5. Solve by factoring, and list every solution: $x^2 + 2x - 15 = 0$

Solution: Look for two numbers with product -15 and sum $+2$. Because the product is negative the numbers have opposite signs, and the larger of the two must carry the sign of the sum: $+5$ and -3 .

$$\begin{aligned}x^2 + 2x - 15 &= 0 \\(x + 5)(x - 3) &= 0\end{aligned}$$

Getting $(x + 3)(x - 5)$ instead reverses both signs and gives $x = -3$ or $x = 5$, which fails the check: $(-3)^2 + 2(-3) - 15 = -12 \neq 0$.

$$\boxed{x = -5 \quad \text{or} \quad x = 3}$$

6. Solve with the quadratic formula, and list every solution: $x^2 - 6x + 5 = 0$

Solution: Here $a = 1$, $b = -6$, $c = 5$. The formula opens with $-b$, and $-b = -(-6) = +6$ — writing -6 there is the sign error this problem is built around.

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} = \frac{6 \pm \sqrt{36 - 20}}{2} = \frac{6 \pm 4}{2}$$

$$\boxed{x = 1 \quad \text{or} \quad x = 5}$$

7. Use the quadratic formula on $2x^2 - 7x + 2 = 0$. Give the larger solution, rounded to two decimal places.

Solution: $a = 2$, $b = -7$, $c = 2$, so $-b = +7$ and the denominator is $2a = 4$.

$$x = \frac{7 \pm \sqrt{(-7)^2 - 4(2)(2)}}{2(2)} = \frac{7 \pm \sqrt{33}}{4}$$

The larger root takes the plus sign: $(7 + 5.7446)/4 = 3.1861\dots$, which rounds to **3.19**

8. Nadia solves $2x^2 + 5x - 3 = 0$ with the quadratic formula, writes the denominator as 2 instead of $2a$, and reports $x = 1$ and $x = -6$. (a) Name the error. (b) Write the correct solution set. (c) Explain how substituting one of her answers would have shown her that something was wrong.

(a) — instructor-judged. The denominator of the quadratic formula is $2a$, not 2. With $a = 2$ it should be 4, so every value Nadia produced is twice as far from zero as it should be. Full credit requires naming the denominator specifically.

Solution (b): $a = 2$, $b = 5$, $c = -3$.

$$x = \frac{-5 \pm \sqrt{25 + 24}}{4} = \frac{-5 \pm 7}{4}$$

$$\boxed{x = -3 \quad \text{or} \quad x = \frac{1}{2}}$$

(c) — instructor-judged. A full answer describes substitution and carries it out on at least one of Nadia's values. Model response: put $x = 1$ into the original equation — $2(1)^2 + 5(1) - 3 = 4$, not 0, so $x = 1$ cannot be a solution and the work must be re-checked. A general remark about “checking your answers”, with no substitution shown, is not full credit.